

What Low-Resource Languages Lose in Dominant-Language Embedding Spaces and How Linguistics Help?

ABSTRACT

Large language models (LLMs) are increasingly mediating translation and language technology, yet their training data is highly uneven: a small set of high-resource languages dominates the field, while many other languages remain underrepresented. Current research deals with this imbalance mainly through technical solutions such as synthetic data generation and few-shot prompting, which can improve performance in low-resource settings. This essay argues that these approaches are necessary but incomplete because they often treat cross-lingual embedding alignment as a neutral step. When low-resource languages are mapped into embedding spaces shaped by the data that came from a different language, specific distinctions from the target language can be compressed even when outputs remain fluent. Using examples from other tongues, this paper shows how translation and alignment can preserve the core message while weakening representational integrity. Finally, the essay argues that linguists are essential to low-resource NLP: they can identify which distinctions matter, guide data and annotation choices and develop evaluations that go beyond surface overlap to test whether models preserve specific ways of encoding knowledge.

INTRODUCTION

Large language models (LLMs) are trained on very large collections of text, which enables them to generate, translate and summarise. Yet the data they learn from is not evenly distributed: a small group of high-resource languages (especially English) accounts for most of the available digital texts, while thousands of other languages have far less representation due to lack of interest or to the small number of speakers (either native or foreign). This imbalance creates a division between high- and low-resource

languages, particularly in the availability of digitised corpora, shared benchmarks and consistent research attention (Joshi et al. ACL 2020).

As a result, building reliable NLP systems for low-resource languages is typically considered a stepped task:

1. Create more data (including synthetic data).
2. Adapt pretrained models efficiently.
3. Use transfer learning so that knowledge learned in dominant languages can generalise to others.

This essay argues that this technical framing is necessary but incomplete. Even when low-resource languages are “supported” through cross-lingual transfer, they are often represented inside semantic spaces shaped mainly by dominant-language distributions. Since languages encode meaning differently, their mapping into the representation of a different language (in this case English) can lose distinctions that carry information related to the culture. In other words, multilingual modelling can increase coverage while still weakening representation integrity: some forms of knowledge become less legible inside the model before any downstream task (translation, summarisation, classification) is even attempted.

The structure of this essay is: first, the literature review outlines current approaches to low-resource NLP as well as parallel issues in low-resource programming languages and concerns about interpretation. It then identifies a gap in how this literature explains meaning loss: cultural and interpretive risks are discussed, but the representational mechanism (dominant-language embedding spaces and their effects) receives less direct attention. Finally, the discussion proposes a new idea, based on linguistic evidence from multiple languages, about what kinds of meaning are most likely to be flattened through dominant-language embeddings and why linguistics is essential for diagnosing and mitigating this loss.

LITERARY REVIEW

Research on low-resource languages in the LLM era is focused on a pragmatic goal: make modern NLP work where data is scarce. Authors repeatedly frame low-resource settings as a gap between:

- (i) the scale and distribution of training data available for dominant languages.
- (ii) the limited digital material, tooling and evaluation resources for many other languages.

For example, Zhong et al. (2026, p. 12) consider low-resource languages as repositories of cultural knowledge, but constrained by scarce datasets, a limited computational infrastructure and technical bottlenecks such as inefficient tokenization for non-Latin scripts or heavy reliance on machine-translated data.

Synthetic data generation to compensate for scarcity

Anikina et al. (2025, p. 2) argue that different methods to prompt synthetic-data are widely used, for example, by putting example outputs right inside the prompt (Cegin et al., 2024a), using descriptions of the labels of data to be generated in the generation prompt (Gu et al., 2024; Abraham et al., 2025) or using self-revision of the LLMs after generation to filter generated data (Li et al., 2024). However, two essential gaps remain. First, these generation strategies are often evaluated separately, and their combinations are not systematically compared, which leaves the overall potential of LLM-based data augmentation unclear. Second, they have been applied to generating English text, leaving aside many other languages (Anikina et al., 2025, p. 2).

However, they affirm that we still lack clear guidance on which strategies (and combinations) work best in low-resource languages. Across conditions, their main result is that “target-language demonstrations plus LLM-based revision” is generally the most reliable combination and even a small number of target-language demos improves downstream quality compared to English demos (Anikina et al., 2025, p. 10). They also repeat that idea with the term “revision”: to filter samples that are ungrammatical, in the wrong output language or irrelevant to the intent, which is especially common in low-resource settings. (Anikina et al., 2025, p. 11).

Concerning machine translation (MT), De Gibert et al. (2025) expand synthetic-data MT by forward-translating English, applying post-processing filters and then pivoting the resulting corpus to build broader multilingual coverage (p. 3).

They assess quality in two ways: automatic checks and human ratings. Overall, the quality of a large part of the synthetic dataset is judged to be high, although the level of quality varies across languages (p. 5). More importantly, the data works well in practice: models trained only on the synthetic corpus can perform almost as well as much larger systems and fine-tuning pretrained baselines gives steady improvements, with the biggest gains in the lowest-resource settings (De Gibert et al., 2025, p. 7).

Based on these experiments, they propose to prioritize volume over perfection for truly low-resource conditions; and, as for adaptation, fine-tune multilingual MT models rather than relying on LLMs for general-purposes (Gibert et al., 2025 p. 8).

Jailbreaking

Jailbreaking also reveals a less discussed consequence of linguistic inequality in LLMs: safety is not distributed equally across languages. Because most safety training and filtering is developed and tested heavily in high-resource languages, protections can be weaker when users switch into low-resource languages or mix several languages in the same prompt (Yong et al. 2023). In this sense, multilinguality is not only a question of performance and access, but also of risk: when safety mechanisms fail more easily outside dominant languages, low-resource communities may face greater exposure to harmful content and misuse.

Evidence suggests that the success rate of such attacks increases substantially when prompts combine multiple low-resource languages, reaching approximately 79% in some cases (Yong et al. 2023). By contrast, high- and mid-resource languages appear to be considerably better safeguarded.

Efficient adaptation of pretrained models

A second common idea is to adapt big pretrained models in an efficient way, instead of training separate monolingual models from scratch, so that low-resource machine translation can work even with limited computing power.

Su et al. (2024, p. 6) show that several PEFT¹ methods perform better than full fine-tuning on both in-domain and out-of-domain tests. They also report the performance gains together with training time, so their claim that PEFT is “efficient” is supported by measured results, not just an assumption (Su et al., 2024, pp. 4–5). However, they also note that the results are not the same in every case: performance differs across language pairs and generally improves when more training data is available, and they discuss how language similarity and resource levels influence the outcomes.

Toraman complements this by studying how to adapt an English-dominant open-source generative LLM to Turkish. They test several adaptation methods, such as continual training, fine-tuning for instructions and specific tasks and vocabulary extension; and try combinations of these approaches (Toraman, 2024, pp. 3–4). The motivation is practical: the tokenizer is strongly biased toward English (“The majority of tokens in its vocabulary are English. The remaining small portion involves European languages with Latin and Cyrillic symbols.”) and expanding it with Turkish tokens increases considerably the number of parameters, so efficiency becomes a key constraint (Toraman, 2024, p. 4).

Toraman also looks at scaling by comparing models with different parameters and testing a multilingual variant (MaLA). The results suggest that larger models can help in some tasks, but adapting the multilingual model does not lead to overall improvements compared to the monolingual-adapted approach (Toraman, 2024, pp. 8–9).

Few-shot and in-context learning for cross-lingual tasks

A third line of research looks at few-shot prompting as a way to reduce the dependency on large parallel datasets. Park et al. (2024, p. 5) test few-shot cross-lingual summarization with GPT-3.5, GPT-4 and an open-source model (Mistral). They find that GPT models usually improve when more examples are added and the biggest improvements appear in many-to-one settings (summarizing into English), where scores are generally higher than in one-to-many tasks (English → low-resource language).

¹Parameter-Efficient Fine-Tuning

Finally, they point out a limitation in evaluation: they use ROUGE (overlap-based) as the only metric, so their results mainly reflect text overlap and cannot fully capture whether the meaning transfer is deeply accurate (Park et al., 2024, p. 6).

Low resources in programming languages

Interestingly, the “low resource” issue does not only happen with human languages. Joel et al. (2024, pp. 51–52) review LLM-based code generation for low-resource programming languages (LRPLs) and domain-specific languages (DSLs).

They explain that the problem does not emerge because of a lack of data, but because these languages often have special syntax and semantics and harder evaluation needs, which also slows progress. They point out that many programming languages need domain-specific metrics and special ways to run or test code, so researchers often use custom evaluation tools instead of the general metrics used for high-resource programming languages (Joel et al., 2024, pp. 18, 51).

They also note a reproducibility problem: since many programming languages do not have standard form, researchers build their own datasets, which makes it harder to compare results across papers and increases the need for shared standards and domain expertise (Joel et al., 2024, p. 52).

This problem is very similar to what happens with low-resource human languages. In both cases, the issue is not only that there is less data, but that the “ecosystem” around the language is thinner: fewer standardized benchmarks or shared evaluation practices and more reliance on task- or community-specific tools.

For human languages, this appears as limited high-quality corpora, uneven coverage in shared test sets and evaluation metrics that may not reflect what matters in real use (for example, culturally specific meaning or pragmatic nuance). As a result, progress can become hard to compare across studies, because researchers often build planned datasets or evaluation setups. This parallel suggests that “low resource” should be treated not simply as a data shortage, but as an infrastructure problem: what languages lack is not only text, but also stable standards, tools and community resources that make development and evaluation cumulative.

Emerging concern: cultural meaning, implicit knowledge and ethics

Finally, a smaller but important strand looks at cultural sensitivity and the limits of LLMs in humanities work. Zhong et al. affirm that models may fail in culturally rich contexts because elements like slang, proverbs, traditional expressions, metaphors, humour and politeness carry local cultural meaning, so general-purpose systems can misunderstand them or simplify them too much (Zhong et al., 2026, p. 12).

They also argue that in literary and religious texts, meaning often depends on hidden or layered references and on cultural or historical knowledge that cannot be fully recovered from the words alone, which increases the risk of overly simplistic interpretations (Zhong et al., 2026, p. 16).

To address this, they call for curated datasets supported by domain expertise (history, philosophy, cultural studies) and for interdisciplinary collaboration, so that surface language is linked to the interpretive frameworks needed for reliable humanities analysis (Zhong et al., 2026, p. 16).

However, these discussions rarely explain how this loss is produced at the representation level or how dominant-language embedding spaces reshape distinctions before any humanities interpretation even happens.

DISCUSSION

As it has been discussed before, most research on low-resource languages in NLP treats the issue as an engineering challenge: collect more data, adapt large models efficiently and use transfer learning so that knowledge from high-resource languages can be applied to others. This work is valuable, but it often treats cross-lingual embedding alignment (mapping different languages into a shared representation space so that tasks like translation become possible) as a technical step.

Embedding alignment can compress language-specific distinctions before any downstream task is attempted. However, as linguists would know, it is important to remember that this mapping is not neutral: if the shared space is shaped mostly by dominant languages (for example, English), then modelling a low-resource language

inside it can flatten or compress some of that language’s specific kinds of knowledge. To follow this way would be like trying to fit an object inside a space that was originally designed for a different one (regardless of how similar their respective shapes are).

One way to understand this is to start from a simple fact about translation. Usually the basic meaning is preserved, but some details are lost or reshaped because languages divide meaning in different ways. They also force different choices about what must be said explicitly, what can be left implicit and which distinctions matter.

One example is politeness and social indexicality. For example, in Japanese, everyday interaction often requires choosing between forms that encode social distance and status (plain vs. polite style; honorific vs. humble choices). English can approximate this with vocabulary (“please”), avoiding a topic or formality, while French and Spanish can with specific pronouns that show politeness (“vous” and “usted/ustedes”²), but they do not require speakers to encode these distinctions grammatically in the same way. When cross-lingual embeddings are trained to align Japanese expressions with English “equivalents”, the model is incentivized to treat sentences that differ crucially in social meaning as close neighbours in vector space. The result can be fluent output that nonetheless loses how the speaker positions themselves socially, a pragmatic dimension that is central in Japanese but weaker in English.

As an example, here is a table for the different ways of saying “to go” in Japanese, with their Romaji reading (the use of the Western alphabet for Japanese words) and in which contexts each word is used:

Verb (kanji/kana)	Romaji	When it is used
行く	iku	Informal.
行きます	ikimasu	The conjugated form of 行く . Everyday standard polite speech.
いらっしゃいます	irasshaimasu	Honorific, to talk about what a respected person (boss/ professor/ customer) does.

² These are singular and plural respectively.

おいでになります	oide ni narimasu	Honorific: similar to いらっしやいます; slightly more “host/service” tone.
行かれます	ikaremasu	Honorific: used on reports and formal narration.
参ります	mairimasu	Humble: Only for 1st person in formal contexts.
行かせていただきます	ikasete itadaki-masu	Polite formula meaning “I shall go (with your permission)”.

In these situations, the different words would probably be translated as “to go”, but the respect or humility that the verb give would probably not be kept.

A second example is subject omission and viewpoint. Japanese and Spanish routinely omit subjects when they are contextually inferable (“行ってきた”³, “Ya llegué”⁴), whereas English and French often force an explicit subject (“I went/I arrived”, “Je suis arrivé(e)”⁵). This is not only a style choice: it affects how a sentence shows agency, shared context and perspective. When systems align languages in a shared embedding space, these subject-free sentences might be matched to English-like versions with explicit pronouns if measures are not taken into account. This can be risky if the context is missing, because the model may guess the wrong subject. Over time, the shared space may start to treat English-style explicitness as the “normal” way to represent events, which can push outputs toward agent-centred wording even when the original language naturally relies more on context than on explicitly naming the agent.

A third example concerns alienable and inalienable possession. Many languages use to mark whether something is inherently linked to a person (inalienable) or more like an owned, transferable object (alienable). A concrete example of alienable versus

³ This phrase [Romaji: ‘ittekita’] can be translated as “I arrived”, but the subject can change according to the context.

⁴ This phrase would truly be translated “I arrived” since it has the Spanish 1st person conjugation.

⁵ The optional “e” would be attached at the end of the word if the subject of the sentence is female

inalienable possession comes from Tongan (a Polynesian language), which uses different possessive markers depending on the type of relationship between a person and what they possess. For items that are seen as closely bound to the person, such as kinship relations or body parts, Tongan commonly uses an o-class possessive, for example *hoku fa'e* ("my mother") or *hoku nima* ("my hand"). For possessions that are more controllable or transferable, it typically uses an a-class possessive, for example *haku me'akai* ("my food") or *haku ta'avale* ("my car"). In English, all of these are expressed with the same form ("my X"), so the translation preserves the basic reference but drops the grammatical distinction about whether the relationship is inherent/closely linked or more like ownership and control. This makes alienable/inalienable contrasts a clear example of how cross-lingual alignment can compress relational knowledge even when the surface meaning remains understandable. This distinction is also found in languages such as Fijian and Tongan, Mayan, Arawakan (located at the Caribbean and Central American), and other Indigenous languages of the Americas and Africa.

Beyond these languages, many others encode distinctions that English does not require, which makes it easy for those differences to be flattened when languages are aligned in an English-shaped embedding space. For instance, some languages grammatically mark evidentiality, forcing speakers to indicate whether information was witnessed directly or learned indirectly (for example, in Turkish, a form like *gelmiş*⁶ can signal reported knowledge, whereas English usually just says "came" unless the speaker adds extra words like "apparently"). Tagalog and Malayan distinguish inclusive and exclusive "we": a "we (including you)" and a "we (excluding you)", a socially important difference that English (and other languages) compresses into one pronoun. Some Slavic languages (and also Spanish and French) encode aspect in ways that systematically distinguish completed events from ongoing or habitual ones (these would be referred as "Perfect Past" and "Imperfect Past" respectively), distinctions that English often reduces to a general past tense.

⁶ Past tense of the verb "to come".

These cases, and countless more as there are more than 7000 alive languages in the world with both unique and shared features, illustrate the main claim: some specific ways of expressing meaning and social nuance are lost when they are forced into a meaning space shaped mostly by a different language⁷. The problem is not that alignment “destroys meaning”, but that it tends to preserve what is richly represented in the dominant-language corpus and to distort what is structurally important in the other language but weakly encoded (or encoded differently) in the dominant language.

Therefore, the key question is not only how to extend model coverage to more languages, but what kinds of knowledge become “unavailable” or less legible inside the model when low-resource languages are mediated through dominant-language embeddings. Addressing this requires complementing performance-driven solutions (the ones discussed before) with an explicit account of representational integrity: which distinctions a language forces speakers to encode and whether multilingual alignment preserves those distinctions or compresses them into dominant-language defaults.

It is important to point out that the language features above mentioned may have been already solved by the existing LLMs, but the point of their presentation is to make the reader aware of unique ways other languages have in order to express meaning, the speakers’ thoughts and the world around them. Nevertheless, the idea of “lost meaning” should not be treated as a vague complaint about translation, but as a linguistic problem that must be solved with tools for analysis. Linguistics can specify *what* is lost, *where* it is encoded and *why* it disappears when moving across languages or aligning languages inside multilingual models and also, *how* to solve it.

For example, typology and semantics show that languages divide up meaning differently; pragmatics explains how social stance and implicature depend on context rather than literal words; and discourse analysis shows how agency and viewpoint can be managed through omission or information structure. These frameworks make it possible to describe meaning loss in precise terms: which distinctions are grammatical

⁷ Although so far English has been considered as that “model language”, this idea can be applied to any other.

or optional, which rely on shared cultural knowledge, and which are carried by form rather than vocabulary.

A key implication of this argument is that linguists are not secondary to low-resource NLP. Engineers can improve coverage and scores, but only linguistic expertise can specify which distinctions a language forces speakers to encode and which meanings are likely to collapse under dominant-language alignment. Linguists can identify where “equivalence” breaks down, distinguish grammatical meaning from optional inference and explain how pragmatic force depends on context and social norms rather than on words alone. Additionally, they can even build the LLM for their native language along with the help of experts in AI and NLP.

This matters because many current evaluation practices reward surface similarity or task accuracy while ignoring whether models preserve language-specific categories and discourse patterns. In practical terms, linguists can guide dataset design in terms of annotation and contexts, create evaluation suites that test representational integrity and help interpret model failures. Without linguists, low-resource work risks producing systems that sound fluent but systematically underrepresent the kinds of knowledge those languages are built to carry.

This matters for AI because cross-lingual embeddings and transfer learning often assume that meaning can be aligned through statistical similarity, but linguistic theory predicts systematic “mismatches” where no clean equivalence exists. Bringing linguistics into the discussion therefore shifts the debate from simply improving performance metrics to evaluating representational integrity: whether models preserve the language-specific categories that shape how speakers encode knowledge, perspective and social relations.

CONCLUSION

This essay has argued that the “low-resource problem” in NLP is not only a question of data scarcity and engineering constraints, but also a question of representation. Current research has produced valuable technical responses that expand coverage and improve downstream performance. Yet coverage alone does not guarantee

that a language is being modelled on its own terms. When low-resource languages are mapped into cross-lingual embedding spaces shaped by dominant-language data, some language-specific distinctions can be compressed even when output remains fluent.

The discussion illustrated this risk with linguistic features that do not map cleanly into English: politeness and social indexicality (especially relevant in Japanese), subject omission and viewpoint management (common in Japanese and Spanish) and possession (very distinguishable in languages from the Polynesia and the Americas). Further examples reinforce the same point: languages partition meaning differently and alignment in an English-centred semantic space can privilege English-like defaults while weakening distinctions that carry epistemic, social or cultural information.

For this reason, linguists should be treated as core contributors or creators of low-resource NLP rather than as optional collaborators. Linguistic theory and method make it possible to identify what is being lost, to separate grammatical meaning from optional inference and to design evaluations that test representational integrity instead of only surface overlap. In practice, this means linguists can guide corpus construction and annotation priorities, develop targeted test suites for language-specific categories and help interpret model failures that standard metrics overlook. If multilingual AI is to be genuinely inclusive, it must aim not only to generate text in more languages, but to preserve the forms of knowledge those languages are built to express and that requires technical innovation working alongside linguistic expertise, community-led data practices and governance that does not treat English as the default template for meaning.

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